2026 INNODEL · INNOVATION IN OPEN AND DISTANCE LEARNING

Mapping an Emerging Research Field

A Bibliometric Analysis of Generative AI in Online and Distance Education

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CONTEXT

Why map this field now

- GenAI creates content: tutoring, feedback, assessment, instructional design.
- Earlier maps cover either e-learning before GenAI, or ChatGPT alone in general higher education: never both together.

The gap: no multi-database map that considers GenAI and online and distance education together.

RQ1 · Temporal trajectory
How has the research evolved over time?

RQ2 · Scientific impact
Which journals and papers drive visibility?

RQ3 · Conceptual structure
Which themes and keyword links define it?

RQ4 · Social organization
How is the community structured?

DATA AND METHOD

Three databases, one consolidated corpus

Web of Science
25
full string · English

Scopus
102
full string · English

ScienceDirect
636
reduced string (8-operator limit)

After DOI merge
735
763 retrieved, 28 duplicates

215
distinct journals

2,725
distinct authors

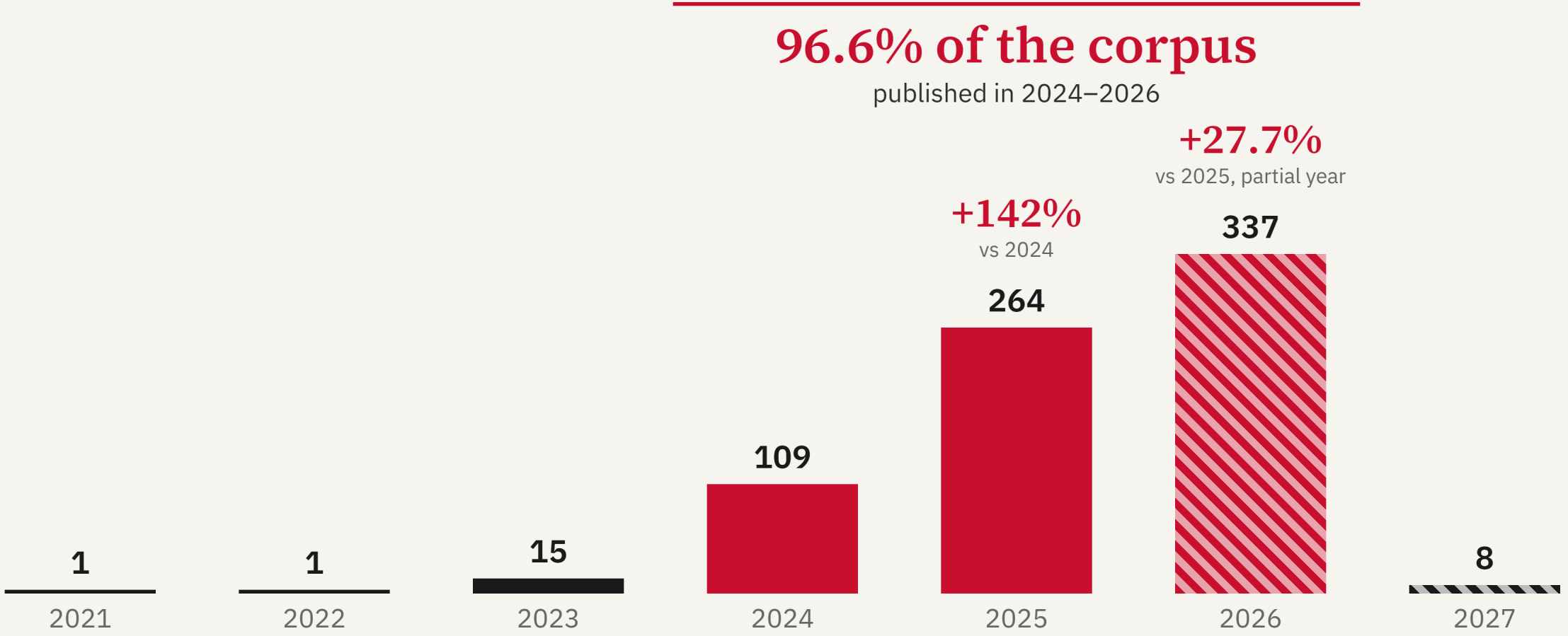
2021–2027
year range

7 Sep 2026
search date

Coverage is uneven. Title, journal and year: 100%. Authors 99.6%, author keywords 97.7%. **Citation counts: only 14.7%**, so citation results use the records that carry them.

RQ1 · TEMPORAL TRAJECTORY

Publications per year



Hatched bars: partial years (search on 7 Sep 2026; 2027 = volumes already indexed).

Journal volume and citation impact

Moderate volume, top impact

British Journal of Educational Technology

6	documents
383	citations
63.8	citations / paper

Clear outlier in the whole corpus.

One paper, 226 citations

Education and Information Technologies

1	document
226	citations
226.0	citations / paper

The journal's rank rests on a single paper.

Top volume, low ratio

Computers and Education: Artificial Intelligence

59	documents
48	citations
0.8	citations / paper

Most productive venue, 8.0% of the corpus.

The 15 most-cited journals hold 92 documents (12.5%) and 1,093 citations; the top two alone hold 55.7% of them. Volume does not predict visibility.

RQ2 · SCIENTIFIC IMPACT

Most-cited papers

British Journal of Educational Technology

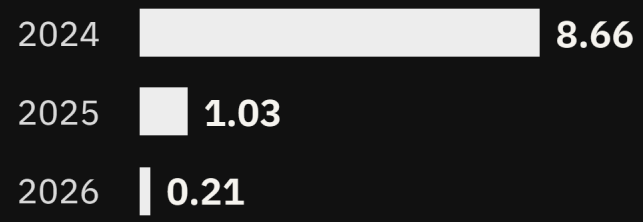
6 papers · 383 citations

319	Empowering student self-regulated learning and science education through ChatGPT: A pioneering pilot study (2024)
21	Aligning and comparing values of ChatGPT and human as learning facilitators: A value-sensitive design approach (2025)
20	Effects of AI-generated adaptive feedback on statistical skills and interest in statistics: A field experiment in higher education (2025)
17	Supporting learner agency in collaborative writing with generative AI (2026)
6	Transforming pedagogy with GenAI-supported formative assessment: Challenges for teacher education (2026)
0	When machines join the moral circle: The persona effect of generative AI agents in collaborative reasoning (2026)

83% of the journal's citations come from a single paper.

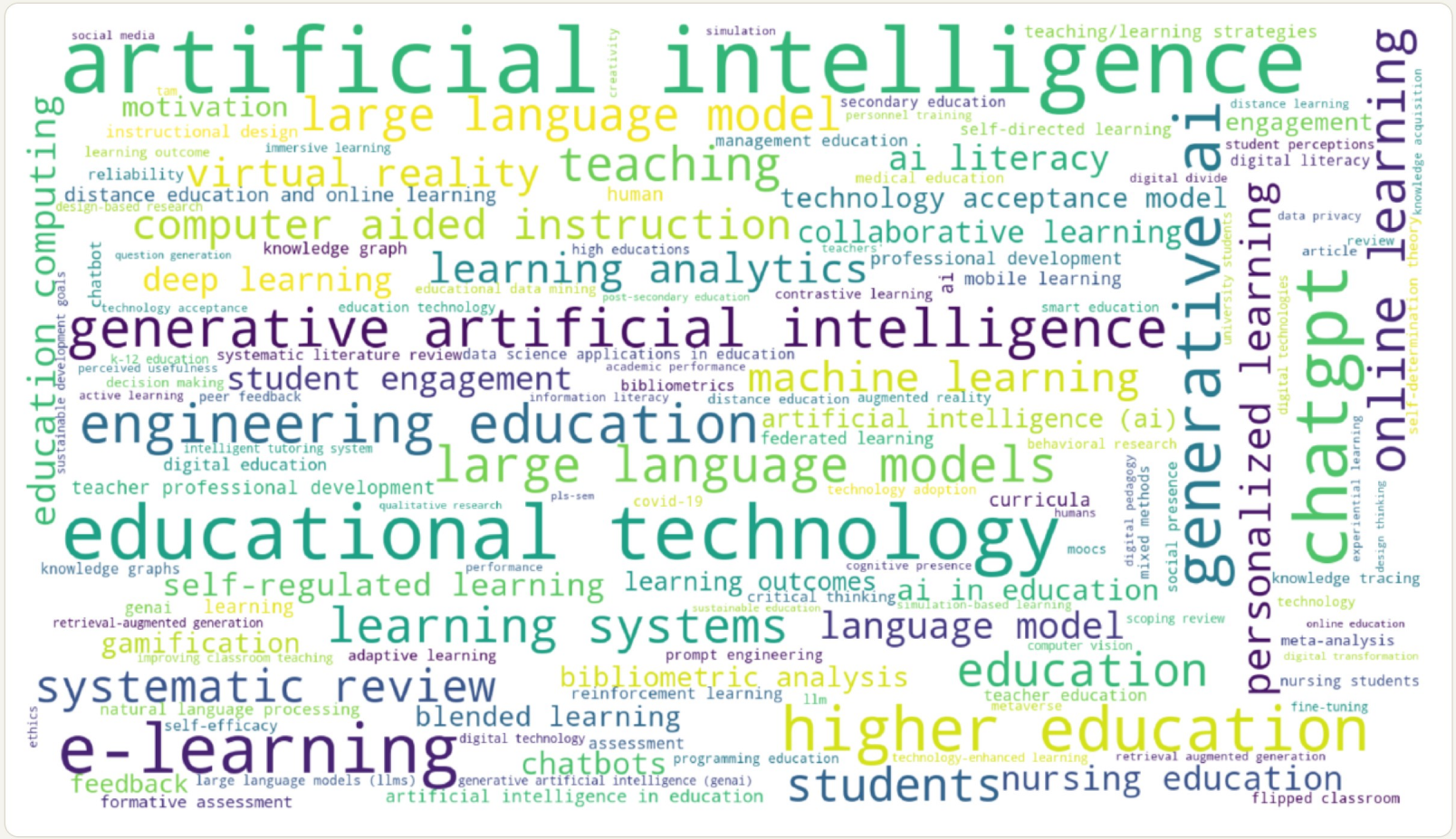
226	Incorporating AI in foreign language education: An investigation into ChatGPT's effect on foreign language learners Education and Information Technologies · the journal's only paper

Average citations per paper, by publication year



2021–2023 papers: negligible rates. Recent papers have had little time to be cited.

Most frequent keywords



Settings

e-learning, online learning, higher education, engineering education

Technology

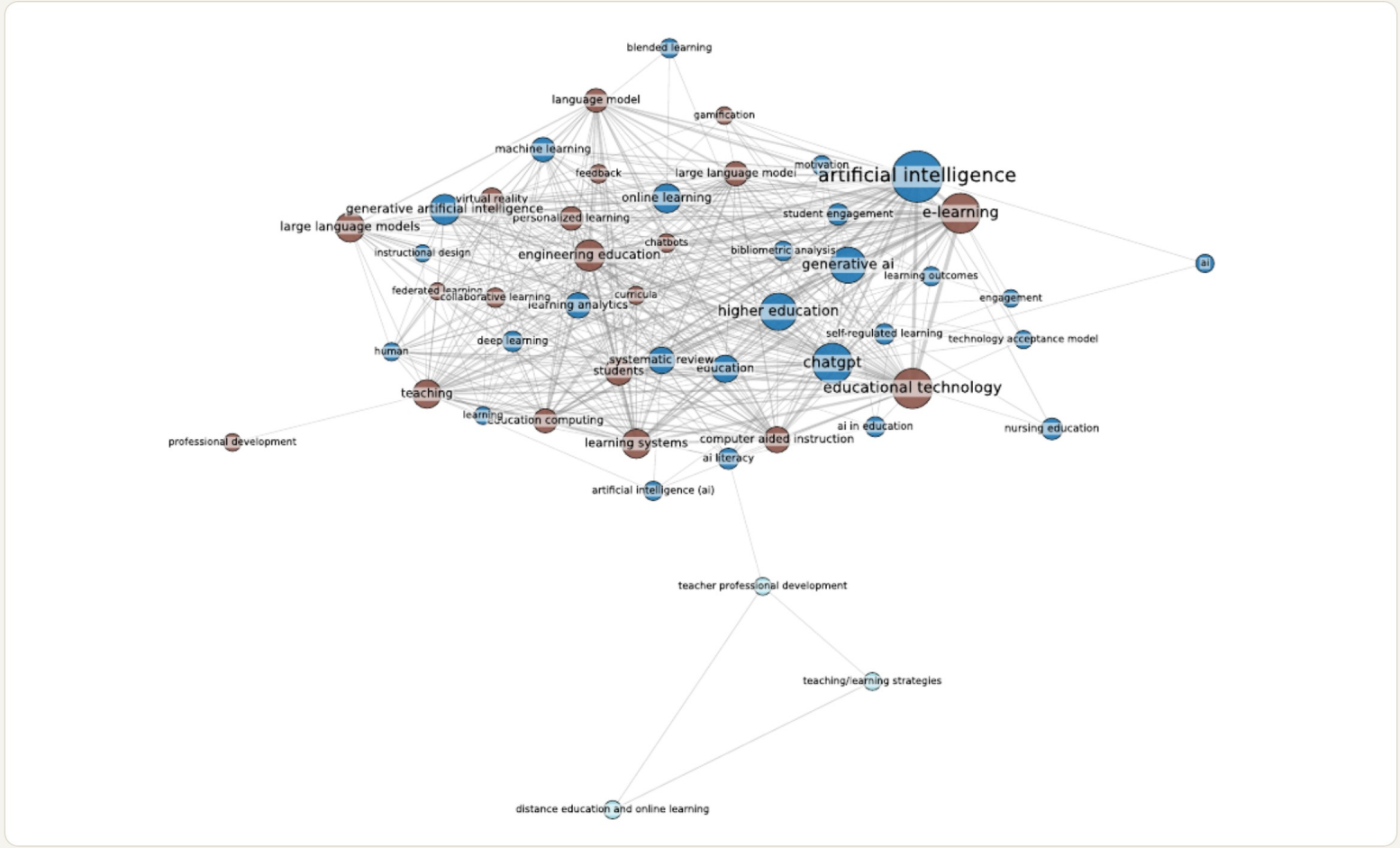
ChatGPT, large language models, machine and deep learning

Pedagogy

teaching, students, learning systems, self-regulated learning

150 most frequent author keywords, out of 2,773 distinct terms.

Keyword co-occurrence network



Strongest links

Documents in which both keywords appear

44 e-learning + educational technology

37 artificial intelligence + e-learning

29 artificial intelligence + ChatGPT

19 ChatGPT + educational technology

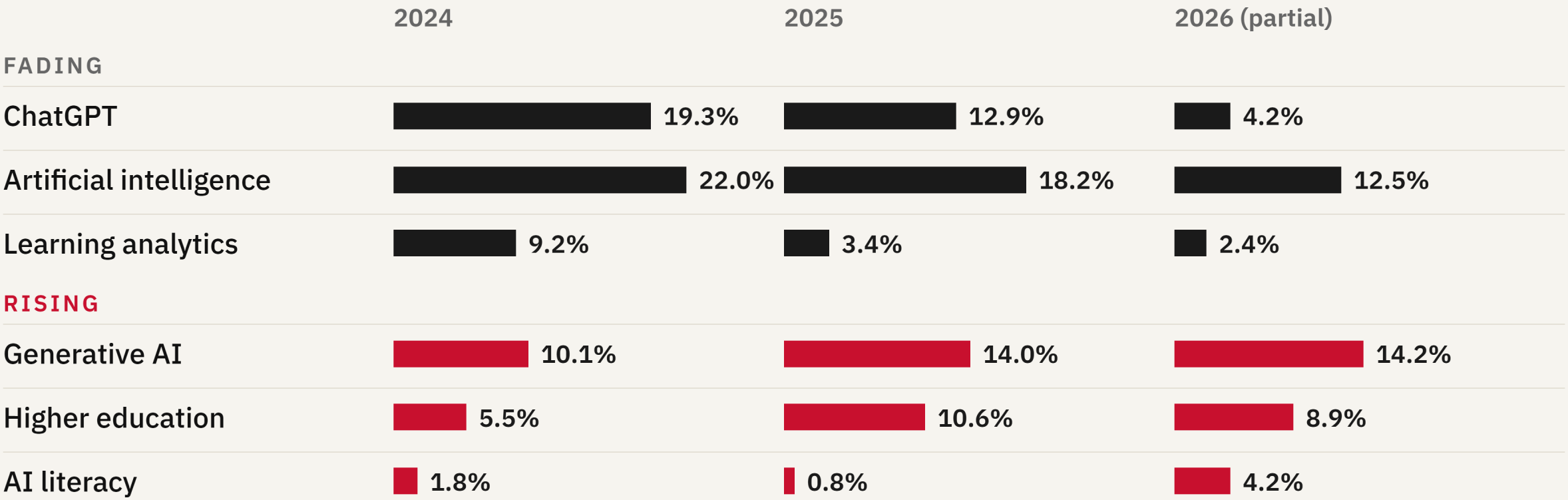
16 ChatGPT + e-learning

13 generative AI + e-learning

Top 50 keywords (of 2,773), minimum 2 co-occurrences.

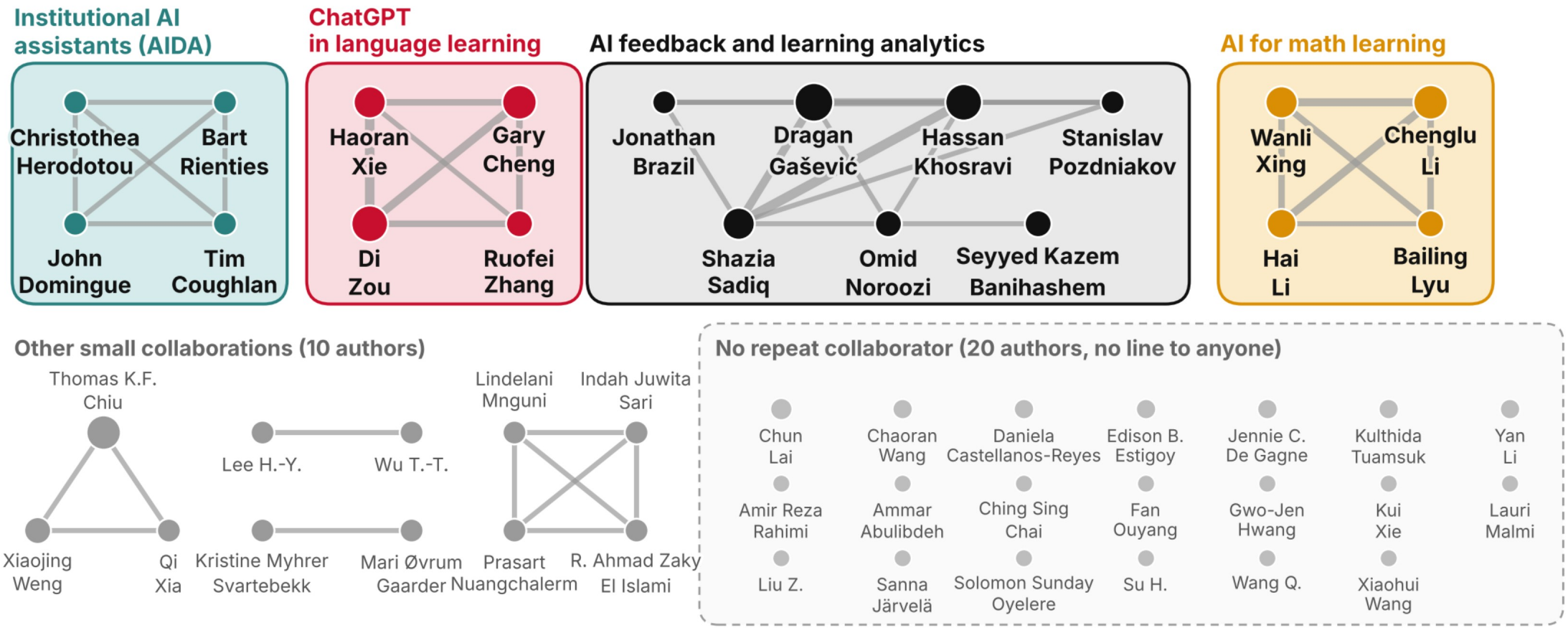
RQ3 · CONCEPTUAL STRUCTURE

From naming the tool to naming the concept



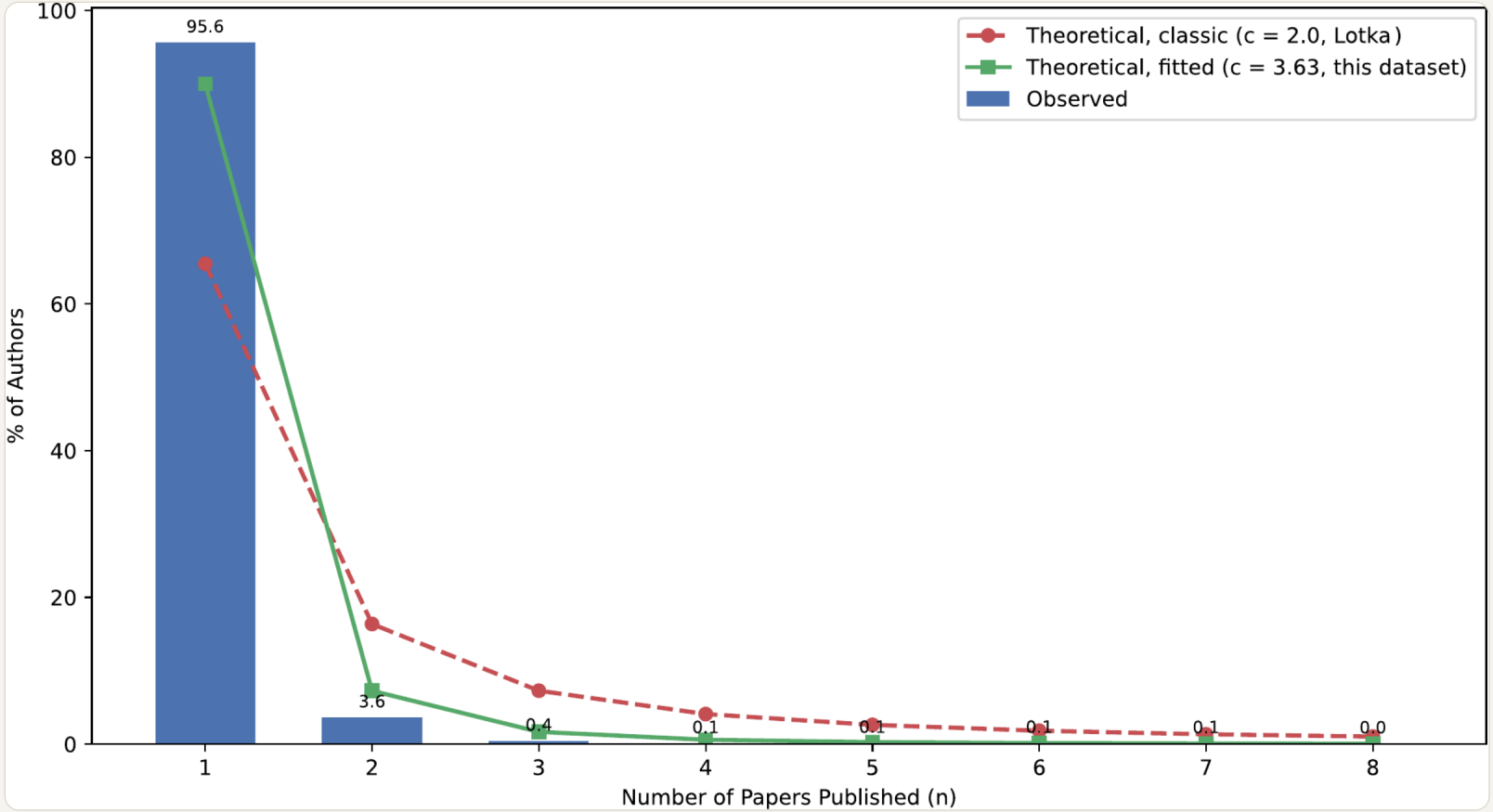
Share of each year's papers using the author keyword (109 papers in 2024, 264 in 2025, 337 in 2026). Spelling variants merged (AI, GenAI). Additional analysis, not in the paper.

Co-authorship: top 50 authors



No group shares 2 or more joint papers with another. Node size = papers in the corpus; line width = joint papers; top 50 authors.

Author productivity



2,725 distinct authors; bars are observed values, lines are Lotka curves.

95.6%

of authors published only one paper

60.8%

expected under the classical Lotka's Law

A few small, tight groups, surrounded by a very transient crowd.

CONCLUSIONS

A field in consolidation

Growth

96.6% of publications appear in 2024–2026, after ChatGPT's release.

Impact

Concentrated in a few landmark papers; volume does not predict citations.

Concepts

Well diversified: LLMs, higher education, e-learning, self-regulated learning.

Community

A few small, tight groups of scholars, but 95.6% of authors publish once.

The conceptual landscape is solidifying faster than the social and collaborative structure.

Limitations

Metadata differs across databases (citations for only 14.7% of records); ScienceDirect search string was reduced; 2026 is a partial year.

Next steps

Tracking how the maps change over time, institutional networks, comparing contexts (levels, disciplines, regions).

Thank you

Questions?

DATA AND ANALYSIS

github.com/Dragonslair5/INNODEL2026

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